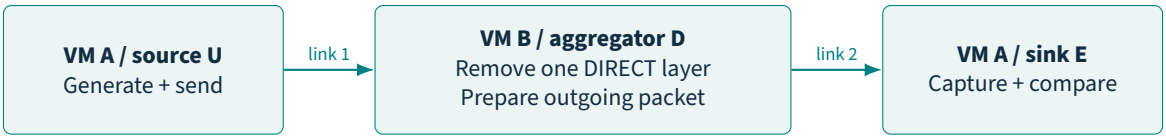
Aggregator forwarding and buffer reuse

The experiment examines whether a PRP aggregator can remove one routing layer and prepare the outgoing packet using existing memory segments, rather than copying its complete body into another assembly buffer. CPU offload means moving work from the main CPU to a device. This test examines a prerequisite for that design, not the device itself; lower elapsed time is not required.

Topology and packet path



The source and sink are separate endpoints in VM A; the aggregator runs in VM B. Two isolated virtual Ethernet links carry the packets. Both guests use one read-only base with private disk overlays; they do not share PRP payload memory. No physical network accelerator is part of this topology.

Protocol elements used in the test

ONION identifies a nested routing representation. Its outer **DIRECT** layer selects the next output and contains a child layer to forward. Removing the outer layer leaves that child intact. The aggregator does not execute the child in this hop.

The **annex** is the attached data region: in this fixture, a short prefix and a deterministic payload. The selected **NONE** session suite does not encrypt the data. Session setup remains authenticated. Annex commitments remain in the packet, but eligible transit does not recompute their hashes in the measured path. Full-byte verification is performed afterwards.

Two outgoing assembly strategies

COPY	Builds the outgoing representation in a contiguous assembly buffer, copying the data that must be forwarded. This is the comparison path.
REFERENCE	Prepares the new header and references retained child/annex segments. The backing memory remains owned and valid until submission completes, avoiding reconstruction of the complete outgoing body at this stage.

Both strategies must emit the **same expected wire bytes**. REFERENCE describes an internal assembly choice, not a new packet format or a reference sent over Ethernet. It does not eliminate every copy: ingress and carrier transmission may still copy data.

Procedure and interpretation

1. Prepare: establish the sessions and route; generate the input and expected output independently. Payload bytes repeat 00–FF.

2. Forward: exercise COPY and REFERENCE in the same selected profile, separating warmup from measurement. Capture the actual input and output.

3. Verify: compare complete delivered bytes with the oracle, bind records to their context and inspect counters for the assembly path. An acknowledgement alone is insufficient.

4. Interpret: preserved delivery plus avoidance of the assembly copy supports this buffer-reuse mechanism. It does not establish NIC offload, CPU savings or physical memory traffic. Such claims require their own measurements and hardware.

The test output supplies the measurements; this note explains the mechanism. The evidence edition can be checked with `./install-verify`; this replays historical observations, not a new VM run. Public native installation remains pending as recorded in RELEASE-STATUS.json.

PRP layer removal and wire representation

Byte ranges below are inclusive, relative to the PRP unit (Ethernet payload). Multi-byte PRP integers are big-endian. $N \in \{256, 512, 1024\}$. Kind 2 means ONION; it is **not** the session cipher-suite field.



Diagram notation. B means bytes; N is the payload length. "Header" groups the 8-byte PRP record prefix and the 2-byte layer length (offsets 0–9), not the Ethernet header. The new header is also 10 bytes: the route alias and lengths change. Arrows indicate unchanged byte ranges. Block widths are schematic.

Input offsets	Output offsets	Bytes / meaning
0–1	0–1	2: record count = 1
2–3	2–3	2: route alias 1 → 2100 (0x0834)
4; 5	4; 5	1 + 1: kind = 2; flags = 0
6–7	6–7	2: body length $N + 100 \rightarrow N + 58$
8–9	8–9	2: DIRECT layer length $84 \rightarrow 42$
10–11	removed	2: outer commitment length = 32
12–43	removed	32: SHA256(annex), precomputed before timing
44–45	removed	2: outer branch count = 1
46–49	removed	4: egress capability = 100
50–51	removed	2: child length = 42
52–53	10–11	2: child commitment length = 32
54–85	12–43	32: child SHA256(annex), unchanged
86–87	44–45	2: child branch count = 1
88–91	46–49	4: child capability = 0xFFFFFFFF
92–93	50–51	2: child length = 0; aggregator does not execute this child
94–95; 96–97	52–53; 54–55	2 + 2: selected annex prefix fields = 17, 3
98–105	56–63	8: sequence = $30 + i$, $i = 0 \ldots 127$ per run
106–107	64–65	2: payload length = N
108–$(N + 107)$	66–$(N + 65)$	N: payload byte $j = j \bmod 256$

Ethernet and local submission envelopes

Ethernet (both links). Bytes 0–5 destination MAC; 6–11 source MAC; 12–13 EtherType `88 FB`; PRP starts at 14. MACs change between links. Captured frame lengths are $N + 122$ in / $N + 80$ out. Captures exclude preamble and FCS.

Netlink submission (tested x86-64 ABI, little-endian). 0–3 length $N + 144$; 4–5 discovered family; 6–7 flags 5; 8–11 sequence; 12–15 port. 16–19: command19/version1/reserved0. 20–23: attribute length12/type23; 24–31: peer64. 32–35: attribute length$N + 112$/type32; 36 onward: input PRP. **ACK36**: 0–15 Netlink header (type2); 16–19 signed error0; 20–35 original request header. An ACK alone is not delivery evidence.

Scope: the selected single-target NONE diagnostic profile, not a general protocol specification. Signed setup/HS2 and other suites are outside this two-page data-wire atlas. Full captured request/ACK/frame bytes and source receipt digests accompany the evidence edition. Integrity hashes bind files; they do not independently authenticate native provenance. No claim of hardware acceleration, complete routing cost or physical zero-copy.